

BEACON LABS

CYBERSECURITY EDUCATION LABS

Workshop



BOISE STATE UNIVERSITY

COLLEGE OF ENGINEERING

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Virtual Machine Setup

Virtual Machine Software Installation

VirtualBox will be the virtual machine software that is used to open the VM images and complete the activity. It is possible to complete the activity with other virtual machine software but it has been untested and so support for other virtual machine software will not be included in this workshop document at this point in time.

Begin by downloading Oracle's VirtualBox software version 7.0.18. The download links for VirtualBox can be found at the below [website](https://www.virtualbox.org/wiki/Download_Old_Builds_7_0). If you already have a newer version of VirtualBox installed, you should be able to follow the same steps.

https://www.virtualbox.org/wiki/Download_Old_Builds_7_0

Windows Setup

Steps:

1. Select the hyperlink titled "Windows hosts" under the "VirtualBox x.x.xx platform packages" as seen in Figure 1 to begin the installation of the latest version of VirtualBox. This activity was created when VirtualBox 7.0.18 was the latest release, however, the steps provided should still apply to future versions of VirtualBox as well.
2. Once the download has completed, run the downloaded installer. The file name should be "VirtualBox-<version_number>-Win.exe".
3. Run through the installer without changing anything. At the end, make sure the checkbox is selected to "Start Oracle VM VirtualBox x.x.xx after installation" as shown in Figure 2 then click the "Finish" button to complete the installation and open VirtualBox.
4. Once VirtualBox has been installed and opened, please continue to the "Downloading and Importing the Images" section.

- **VirtualBox 7.0.18** (released May 07 2024)
 - **Windows hosts**
 - macOS / Intel hosts
 - Solaris hosts
 - Solaris 11 IPS hosts
 - Linux Hosts:
 - Oracle Linux 9 / Red Hat Enterprise Linux 9
 - Oracle Linux 8 / Red Hat Enterprise Linux 8
 - Oracle Linux 7 / Red Hat Enterprise Linux 7 / CentOS 7
 - Ubuntu 24.04
 - Ubuntu 22.04
 - Ubuntu 20.04
 - Ubuntu 18.04 / 18.10 / 19.04
 - Debian 12
 - Debian 11
 - Debian 10
 - openSUSE 15.3 / 15.4 / 15.5
 - Fedora 40 / 41 / 42
 - Fedora 36 / 37 / 38 / 39
 - Fedora 35
 - All distributions
 - Extension Pack
 - Guest Additions
 - Sources
 - MD5 checksums, SHA256 checksums

Figure 1: Download button is highlighted in gray for Windows machines.



Figure 2: Final screen for the Windows installer.

Linux Setup

1. From the VirtualBox download page seen in Figure 1, find your distribution of Linux and follow the instructions Oracle has provided.
2. If your distribution isn't shown on the website, please look into distribution specific instructions for your Linux distro. For example, if you are using Arch Linux, there is a VirtualBox page on the ArchWiki that provides an installation guide.

Downloading and Importing the Images

This activity in the workshop will be done using a single virtual machine that will be provided to you. Said image can be downloaded from [this Google Drive link](https://drive.google.com/file/d/1FWWNZCXi8gSi9SsbiBaMJNuyTAFvbDps/view?usp=sharing).

<https://drive.google.com/file/d/1FWWNZCXi8gSi9SsbiBaMJNuyTAFvbDps/view?usp=sharing>

Steps:

1. Use the above link to open Google Drive to two virtual machine image file.
2. Download the image file. The file is 5.54 GB in size so the download may take a while.
3. Once the images has been downloaded, it can be imported into VirtualBox by navigating to File > Import Appliance and selecting the .ova image file that was downloaded. Visual instructions can be seen in Figures 3, 4, 5, 6, and 7.

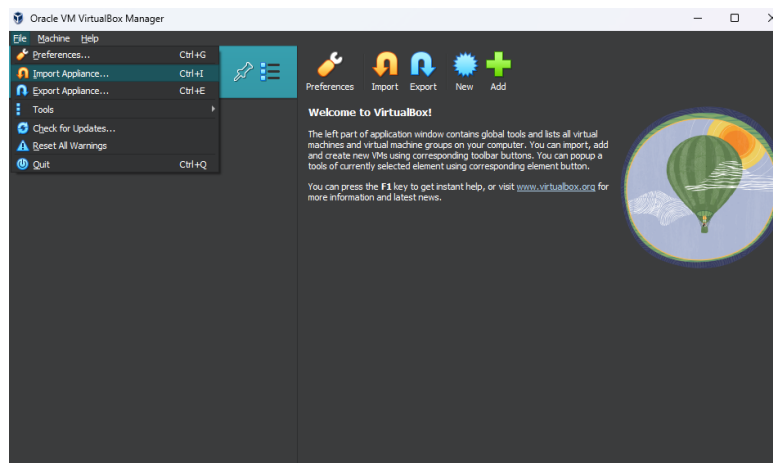


Figure 3: File > Import Appliance

4. Once the image has been imported to VirtualBox, continue to the final section, "Verifying Import."

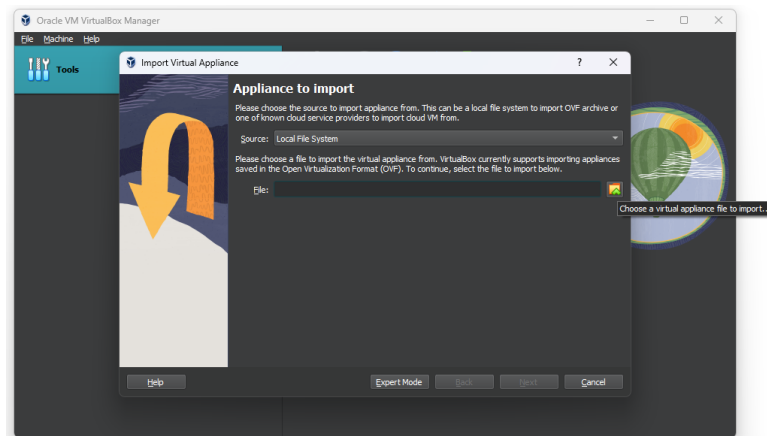


Figure 4: Select the file button to choose the downloaded image file.

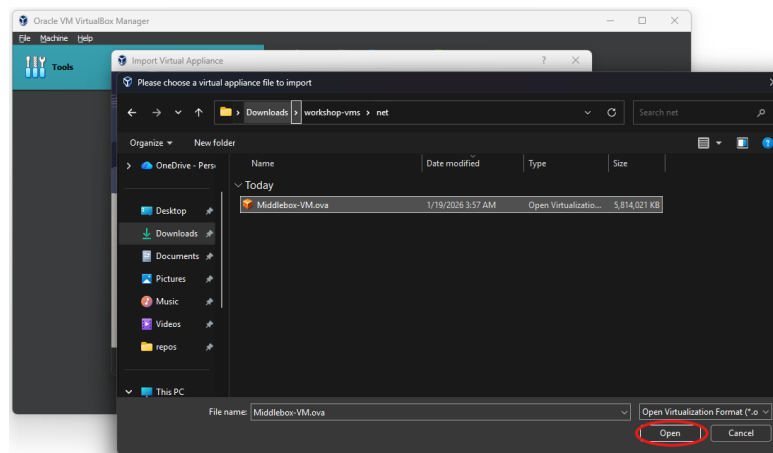


Figure 5: Choose the downloaded .ova file to import.

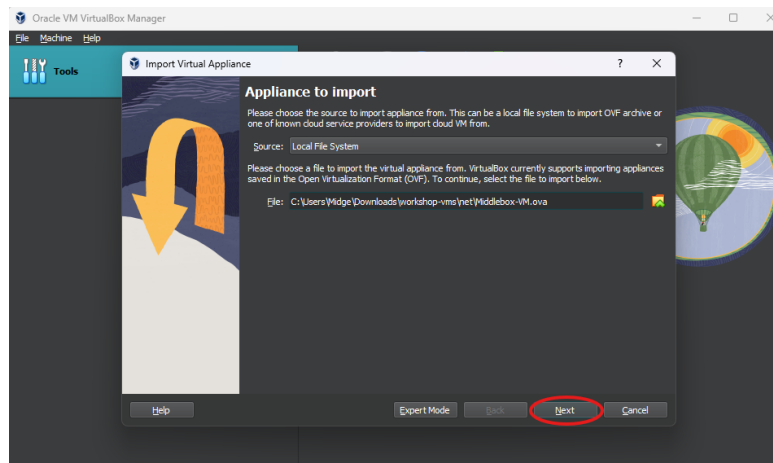


Figure 6: Select next after selecting the image to import.

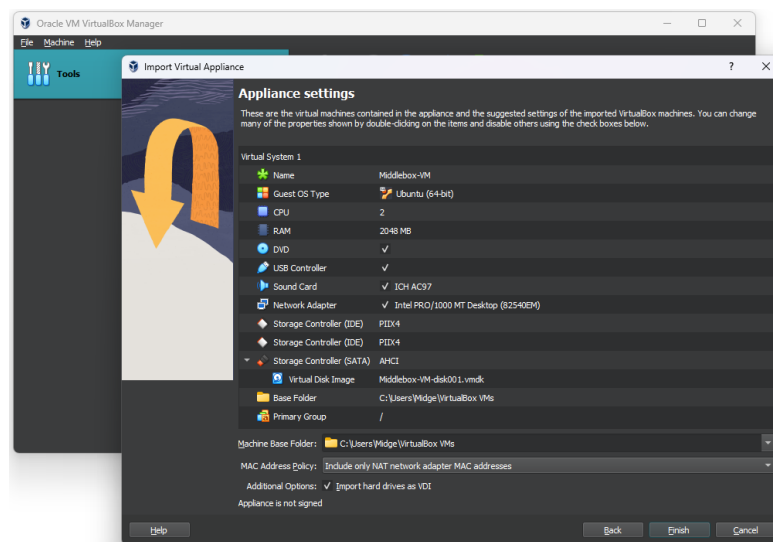


Figure 7: Final confirmation. Your file paths and names may differ slightly.

Verifying Import

Steps:

1. Start the virtual machine.
2. Use “middlebox” as the username and password to log in.
3. Open a terminal by either selecting the terminal application on the Desktop or App Launcher or executing the keyboard shortcut Ctrl+t.
4. Verify that Java can be found and run by executing the command `java -version`.
5. If Java was found, you are done, continue to Step 7. If not, continue to the next Step.
6. Download Java by executing the command `sudo apt-get install openjdk-17-jdk`. Use “middlebox” as the password if prompted. After the download is complete, repeat Step 4.
7. Congratulations, you’ve successfully imported the virtual machine image and are now ready for the Network Security workshop segment of the BEACON Labs Workshop. Thank you for coming prepared! If you haven’t already, please follow the instructions for the Web and Mobile Security material setups. We look forward to seeing you in Boise for the workshop!